

2.Chebyshev IIR Filters 2.1 Design and performance Evaluation of a Chebyshev Type I IIR Low Pass Filter using MATLAB

Programming Code:

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clc;
clear;
close all;

%% Filter Specifications
Fs = 1000;    % Sampling Frequency (Hz)
Fp = 100;     % Passband Frequency (Hz)
Fst = 150;    % Stopband Frequency (Hz)
Rp = 1;       % Passband Ripple (dB)
Rs = 40;      % Stopband Attenuation (dB)

%% Normalize Frequencies
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

%% Calculate Minimum Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

%% Design Chebyshev Type-I Low Pass Filter
[b,a] = cheby1(n,Rp,Wn,'low');

%% Display Filter Information
disp('Chebyshev Type-I IIR Low Pass Filter');
disp(['Filter Order = ',num2str(n)]);
disp('Numerator Coefficients (b):');
disp(b);
disp('Denominator Coefficients (a):');
disp(a);

%% Frequency Response
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figure;
freqz(b,a,1024,Fs);
title('Frequency Response of Chebyshev Type-I Low Pass Filter');

%% Magnitude and Phase Response
[H,f] = freqz(b,a,1024,Fs);
figure;
subplot(2,1,1);
plot(f,20*log10(abs(H)),'b','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response');
subplot(2,1,2);
plot(f,unwrap(angle(H))*180/pi,'r','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response');

%% Impulse Response
figure;
impz(b,a);
grid on;
title('Impulse Response');

%% Pole-Zero Plot
figure;
zplane(b,a);
grid on;

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title('Pole-Zero Plot');

%% Test Signal
t = 0:1/Fs:1;

% Passband Signal
x1 = sin(2*pi*50*t);

% Stopband Signal
x2 = 0.5*sin(2*pi*250*t);

% Combined Signal
x = x1 + x2;

%% Filter Output
y = filter(b,a,x);

%% Time Domain Signals
figure;

subplot(2,1,1);

plot(t,x,'b');

grid on;

xlabel('Time (s)');

ylabel('Amplitude');

title('Input Signal');

subplot(2,1,2);

plot(t,y,'r');

grid on;

xlabel('Time (s)');

ylabel('Amplitude');

title('Filtered Output Signal');

%% FFT Analysis
N = length(x);

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X = abs(fft(x));
Y = abs(fft(y));
faxis = (0:N-1)*(Fs/N);
figure;
subplot(2,1,1);
plot(faxis,X,'b');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Input Signal');
subplot(2,1,2);
plot(faxis,Y,'r');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Filtered Signal');
%% Display Filter Coefficients
fprintf('\nNumerator Coefficients (b):\n');
disp(b);
fprintf('\nDenominator Coefficients (a):\n');
disp(a);

```

OUTPUTS:



